

Joshua Negreanu

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AI Engineer focused on deep multimodal learning, robotics data collection and training, and generative models.

EDUCATION

- Oregon State University**

• MS Artificial Intelligence (*Accelerated*), GPA: 4.00

Relevant Coursework: Deep Learning, NLP with Deep Learning, ML, Online ML, Optimization, Algorithms and Data Structures.

Corvallis, Oregon
September 2024 - June 2026
- Oregon State University**

• BS Computer Science (*AI focus*), GPA: 4.00
BS Mathematics (*Applied focus*), GPA: 4.00

Relevant Coursework: OS I and II, ML, Parallel Programming, Applied PDEs, Systems of ODEs, Computational Number Theory.
- Corvallis, Oregon
September 2021 - June 2025
September 2021 - June 2025

SKILLS SUMMARY

- **Languages:** Python, C/C++, JavaScript, Java, Bash, SQL, Elm
- **ML/Math/Stats:** PyTorch, NumPy, SciPy, Scikit-learn, Wandb, LeRobot
- **Systems/Acceleration:** CUDA, OpenMP, OpenCL, GStreamer
- **Data/Infrastructure:** Pandas, Matplotlib, OpenCV, AWS, Azure, DGX, Jetson
- **Backend:** Node.js

EXPERIENCE

- Revobots**

• AI Intern

- **Remote Teleoperation:** Designed and deployed a mediasoup-based SFU enabling low-latency 30-fps real-time video conferencing streaming of robot vision and control signals for remote teleoperation. Supports 500+ clients per robot.
 - **VR Embodiment:** Implemented real-time 60-fps 4k stereo image streaming and headset-controlled embodiment for dataset recording with WebXR integration and GPU accelerated encoding via GStreamer.
 - **Data Collection:** Utilized Hugging Face LeRobot framework for teleoperation, 100+ episode dataset recording, compression. Automated storage and metadata tagging with AWS.
 - **Model Training:** Employed MLOps for automated multi-day training of transformer-based robot policies on DGX system using collected datasets.

Remote
June 2025 - Present
- Oregon State University**

• Educational Assistant

- **Programming Language Fundamentals GTA:** Hosted office hours, graded assignments on functional programming, language syntax and semantics in Haskell and Prolog, implemented autograder scripts for unit testing.
 - **Operating Systems I ULA:** Held office hours for 40+ students, graded UNIX OS projects spanning multithreading, shell development, file I/O, network protocols.
 - **Data Structures ULA:** Conducted grading demos, office hours for 30+ students, met weekly with professor.
 - **Gen Engineering ULA:** Led recitations with fellow assistants on industry software (COMSOL, MATLAB, Simulink).

Onsite
Sep 2022 - Present

RESEARCH

- **Master's Thesis:** In progress, exploring automated filtering of training data to help generalize/strengthen robot policies. Under the mentorship of Rahul Khanna.
- **Undergraduate Thesis: Attacks on Stochastic Dueling Bandits:** Conceptualized and proposed mathematical framework for reward poisoning algorithms within stochastic dueling bandits context. Encourage development of corruption-robust algorithms. Under the mentorship of Huazheng Wang. Paper available on *OSU Archives*.

PROJECTS

- **Discrete Diffusion and Auto-Regressive Language Model:** From-scratch implementation of both traditional auto-regressive and discrete diffusion generative transformer-based language models. Trained on openwebtext utilizing DGX system. Open-source code available on *GitHub*.
- **Denoising Diffusion and Flow Matching Image Generation:** From-scratch implementation of denoising diffusion and flow matching image generative models (CNN U-Net and latent vision transformer). Trained on Stanford Cars, CelebA, ImageNet utilizing DGX system. Open-source code available on *GitHub*.
- **Math Machine Translation:** Implementation of novel seq2seq transformer model for math word problem to computational graph construction. Trained on MathQA dataset utilizing DGX system. Open-source code and report available on *GitHub*.

HONORS AND AWARDS

- **CV Copson Scholarship:** Lonseth Lecture May 2025
- **WIC Culture of Writing Award:** Lonseth Lecture May 2025
- **Dean's List:** Sep 2021 - June 2025

PERSONAL INTERESTS

- **Sports:** Ultimate Frisbee Club (*Oregon State University*), Varsity Track and Field (*Catlin Gabel*), 2nd Degree Tae Kwon Do (*instructor, demo team*), rock climbing, running, backpacking.
- **Creative:** Competitive Piano (*9 years, OFMC 3rd Gold Cup, Oregon Duet Competition Winner*), guitar, music production.